

Chapter 5

SINGULAR SOLUTIONS

5.1 Introduction

In the chapter 1, we have already introduced the idea of singular solution of a differential equation and in the chapter 4, we have given examples on it. In this chapter, we present a general discussion on singular solution of a differential equation after giving a few definitions related to singular solution.

The *discriminant* of an equation involving single parameter is a function of the coefficients in the rational integral form. For the function $f(x, y, \alpha) = 0$, α being the parameter, the discriminant is the α eliminant of the equations $f(x, y, \alpha) = 0$ and $\frac{\partial f}{\partial \alpha} = 0$.

If $\phi(x, y, c) = 0$ be the general solution of the differential equation $f(x, y, p) = 0$, where $p = \frac{dy}{dx}$. Then p -discriminant relation is obtained by eliminating p from $f(x, y, p) = 0$, $\frac{\partial f}{\partial p} = 0$ and represents the locus, for each point of which $f(x, y, p) = 0$ has equal values of p .

Similarly, the c -discriminant relation is obtained by eliminating c from $\phi(x, y, c) = 0$, $\frac{\partial \phi}{\partial c} = 0$ and represents the locus, for each point of which $\phi(x, y, c) = 0$ has equal values of c .

The equation $f(x, y, p) = 0$ and its solution $\phi(x, y, c) = 0$ are of same degree in p and c respectively, so if there be a p -discriminant relation, then there must be a c discriminant relation.

5.2 Working Rule for Finding Singular Solution

Let $F(x, y) = 0$ be the p -discriminant relation, i.e., the p -eliminant of $f(x, y, p) = 0$ $\frac{\partial f}{\partial p} = 0$. If $F(x, y) = 0$ satisfies the differential equation $f(x, y, p) = 0$, then it is called its singular solution. If not so then the factors of it, which satisfy the equation, will form the singular solution of the equation. Clearly, the equation $f(x, y, p) = 0$ will have no singular solution, if it can be resolved into linear factors in p .

Again, let $g(x, y) = 0$ be the c -discriminant relation, i.e., the c -eliminant of $\varphi(x, y, c) = 0$, $\frac{\partial \varphi}{\partial c} = 0$. If $g(x, y) = 0$ satisfies the differential equation $f(x, y, p) = 0$, then it is called its singular solution. If not so, then the factors of it, which satisfy the equation, will form the singular solution of the equation.

If the equation $f(x, y, p) = 0$ be given in Clairaut's form

$$y = px + h(p),$$

$$y = cx + h(c).$$

then its general solution is

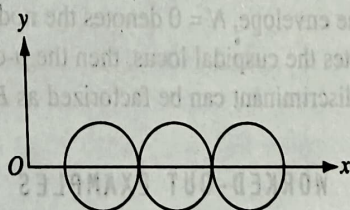
In this case both p -discriminant relation and c -discriminant relation will be same, i.e., both of these will give us the required solution, which is not true for other forms of the equation $f(x, y, p) = 0$.

5.3 Extraneous Loci

If $s(x, y) = 0$ is a singular solution of the differential equation $f(x, y, p) = 0$ whose general solution is $\varphi(x, y, c) = 0$, then $s(x, y)$ is a factor of both p -discriminant and c -discriminant. These, discriminants, however, may have other factors which give rise to other loci associated with the general solution. Since, these equations of the loci do not satisfy the equation $f(x, y, p) = 0$, generally, these are called *extraneous loci*.

Tac locus: Let P be a point on p -discriminant relation, then there are two equal values of p at the point P . If it is found that these equal values of p belong to two distinct, non-consecutive curves touching at P , then this point will satisfy the p -discriminant but not c -discriminant. Again, the point P does not lie on the envelope; hence this will not satisfy the differential equation. Locus of such point is called *tac locus*. If $T(x, y) = 0$ denotes the equation of tac locus, then $T(x, y)$ is a factor of p -discriminant. In general, $T(x, y)$ is not a factor of c -discriminant and $T(x, y) = 0$ does not satisfy the differential equation.

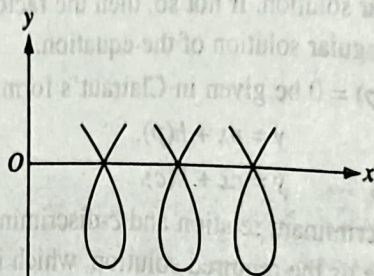
Consider a system of circles of fixed radius and having centres on x -axis. Here, the x -axis is the locus of the point of contacts of the non consecutive circles of the system and represents the tac locus.



$y = 0$ is a tac locus.

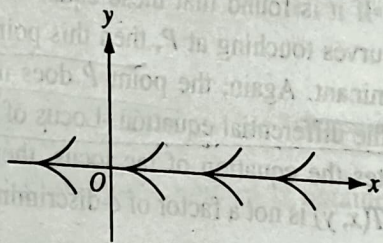
Nodal locus: The c -discriminant relation represents the locus of the points for which two values of c are equal. If each of the curve of the family $\varphi(x, y, c) = 0$ has a node (a double point with distinct tangents), then the locus of the nodes is called *nodal locus*. If $N(x, y) = 0$ denotes the equation of nodal locus, then $N(x, y)$ is a factor of

c -discriminant. In general, $N(x, y)$ is not a factor of p -discriminant and $N(x, y) = 0$ does not satisfy the differential equation.



$y = 0$ is a nodal locus.

Cuspidal locus: If each member of the family of curves represented by the general solution $\varphi(x, y, c) = 0$ of the differential equation $f(x, y, p) = 0$ has a cusp (a double point with coincident tangent), then the locus of the cusps is called *cuspidal locus*. Since, a cusp is also the point of intersection of consecutive curves, both p -discriminant relation and c -discriminant relation will contain the cuspidal locus. If $C(x, y) = 0$ denotes the equation of cuspidal locus, then $C(x, y)$ is a factor of both c -discriminant and p -discriminant. In general, $C(x, y)$ does not satisfy the differential equation.



$y = 0$ is a cuspidal locus.

Thus the p -discriminant relation may contain singular solution, the tac locus and the cuspidal locus whereas the c -discriminant relation may contain singular solution, the nodal locus and the cuspidal locus. Here, we give two important results without proof. If $E = 0$ denotes the envelope, $N = 0$ denotes the nodal locus, $T = 0$ denotes the tac locus and $C = 0$ denotes the cuspidal locus, then the p -discriminant can be factorized as ET^2C and the c -discriminant can be factorized as EN^2C^3 .

WORKED-OUT EXAMPLES

1. Find the singular solution of the equation $p^2 + y^2 - 1 = 0$.

Solution

From the equation we get $\frac{dy}{dx} = \sqrt{1 - y^2}$. Integrating, we get

$$\sin^{-1} y = x + c \quad \text{or, } y = \sin(x + c).$$

This is the general solution.

Using the p -discriminant relation, we get

$$y^2 - 1 = 0 \quad \text{or, } y = \pm 1.$$

These solutions satisfy the given equation. Hence singular solutions of the given equation are $y = \pm 1$.

- ✓ 2. Find the singular solution of the equation $xp^2 - (x - a)^2 = 0$.

Solution

From $xp^2 - (x - a)^2 = 0$, we get

$$p = \sqrt{x} - \frac{a}{\sqrt{x}}.$$

Integrating, we get $y + c = \frac{2}{3}x^{\frac{3}{2}} - 2a\sqrt{x}$

$$\text{or, } (y + c)^2 = \frac{4}{9}x(x - 3a)^2$$

$$\text{or, } c^2 + 2cy + y^2 - \frac{4}{9}x(x - 3a)^2 = 0.$$

So, the c -discriminant is $4y^2 = 4y^2 - \frac{16}{9}x(x - 3a)^2$

$$\text{or, } x(x - 3a)^2 = 0.$$

The p -discriminant is $x(x - a)^2 = 0$.

So, singular solution is $x = 0$, tac locus is $x = a$ and nodal locus is $x = 3a$.

3. Find the singular solutions of the equation $(3y - 1)^2 p^2 = 4y$.

Solution

From $(3y - 1)^2 p^2 = 4y$, we get

$$p = \pm \frac{2\sqrt{y}}{3y - 1}$$

$$\text{or, } \frac{3y - 1}{\sqrt{y}} dy = \pm 2dx$$

$$\text{or, } \left(3\sqrt{y} - \frac{1}{\sqrt{y}} \right) dy = \pm 2dx.$$

Integrating, we get

$$2 \left(y^{\frac{3}{2}} - \sqrt{y} \right) = 2(c \pm x)$$

$$\text{or, } (x \pm c)^2 = y(y - 1)^2$$

$$\text{or, } c^2 \pm 2cx + x^2 - y(y - 1)^2 = 0.$$

So, the c -discriminant is

$$4x^2 = 4x^2 - 4y(y-1)^2 \quad \text{or, } y(y-1)^2 = 0.$$

The p -discriminant is

$$y(3y-1)^2 = 0.$$

So, singular solution is $y = 0$, tac locus is $3y = 1$ and nodal locus is $y = 1$.

4. Find the singular solutions of the equation $x^3 p^2 + x^2 y p + a^3 = 0$.

Solution

From $x^3 p^2 + x^2 y p + a^3 = 0$, we get

$$y = -xp - \frac{a^3}{x^2 p}.$$

Differentiating, we get

$$p = -p - x \frac{dp}{dx} + \frac{2a^3}{x^3 p} + \frac{a^3}{x^2 p^2} \frac{dp}{dx} \quad \text{or, } \left(\frac{dp}{dx} + \frac{2p}{x} \right) \left(\frac{a^3}{x^2 p^2} - x \right) = 0.$$

From

$$\frac{dp}{dx} + \frac{2p}{x} = 0,$$

we get

$$\frac{dp}{p} + 2 \frac{dx}{x} = 0.$$

Integrating, we get

$$px^2 = c.$$

Using this in the given equation, we get

$$c^2 + cxy + a^3 x = 0.$$

The c -discriminant is

$$x^2 y^2 - 4a^3 x = 0 \quad \text{or, } x(xy^2 - 4a^3) = 0.$$

The p -discriminant is

$$x^4 y^2 - 4a^3 x^3 = 0 \quad \text{or, } x^2 x(xy^2 - 4a^3) = 0.$$

Hence, $x(xy^2 - 4a^3) = 0$ is the singular solution and $x = 0$ is also tac locus.

5. Find the singular solutions of the equation $2p^2 - 2x^2 p + 3xy = 0$.

Solution

$$\text{From } 2p^2 - 2x^2 p + 3xy = 0, \text{ we get } y = \frac{2}{3} px - \frac{2p^2}{3x}.$$

$$\text{Differentiating we get } p = \frac{2}{3} p + \frac{2}{3} x \frac{dp}{dx} + \frac{2p^2}{3x^2} - \frac{4p}{3x} \frac{dp}{dx}$$

$$\text{or, } \left(2 \frac{dp}{dx} - \frac{p}{x} \right) \left(x - \frac{2p}{x} \right) = 0.$$

From

$$2 \frac{dp}{dx} - \frac{p}{x} = 0,$$

we get

$$p^2 = cx.$$

Using this in the given equation, we get

$$2cx + 3xy = 2x^2\sqrt{cx}$$

$$\text{or, } (2c + 3y)^2 = 4x^3c$$

$$\text{or, } 4c^2 + 4c(3y - x^3) + 9y^2 = 0.$$

The c -discriminant is

$$16(3y - x^3)^2 - 144y^2 = 0 \text{ or, } x^3(x^3 - 6y) = 0.$$

The p -discriminant is

$$4x^4 - 24xy = 0 \text{ or, } x(x^3 - 6y) = 0.$$

Hence, $x^3 - 6y = 0$ is the singular solution and $x = 0$ is the cuspidal locus.

6. Find the singular solutions of the equation $p^2(1 - x^2) = 1 - y^2$.

Solution

From $p^2(1 - x^2) = 1 - y^2$, we have

$$p = \sqrt{\frac{1 - y^2}{1 - x^2}} \text{ or, } \frac{dy}{\sqrt{1 - y^2}} = \frac{dx}{\sqrt{1 - x^2}}.$$

Integrating, we get

$$\sin^{-1} y - \sin^{-1} x = c'$$

$$\text{or, } \cos^{-1} x - \cos^{-1} y = c'$$

$$\text{or, } \cos^{-1} [xy + \sqrt{1 - x^2}\sqrt{1 - y^2}] = c'$$

$$\text{or, } xy + \sqrt{1 - x^2}\sqrt{1 - y^2} = \cos c' = c$$

$$\text{or, } (1 - x^2)(1 - y^2) = (c - xy)^2$$

$$\text{or, } c^2 - 2xyc + x^2 + y^2 - 1 = 0.$$

So, the c -discriminant is

$$4x^2y^2 = 4(x^2 + y^2 - 1) \text{ or, } (1 - x^2)(1 - y^2) = 0.$$

Also, the p -discriminant is $(1 - x^2)(1 - y^2) = 0$.

Hence, the singular solutions are $x = \pm 1, y = \pm 1$ and it has no extraneous loci.

7. Reduce the differential equation $(y + xp)^2 = x^2p$ to Clairaut's form by the substitution $xy = v$ and then examine for singular solution.

Solution

From $xy = v$, we get $xp + y = \frac{dv}{dx} = P$ (say). Using this in the given equation, we get

$$P^2 = x(P - y) = xP - v \text{ or, } v = xP - P^2.$$

This is in Clairaut's form. So the general solution is

$$v = xc - c^2 \text{ or, } xy = xc - c^2.$$

The c -discriminant is

$$x^2 - 4xy = 0 \text{ or, } x(x - 4y) = 0.$$

The p -discriminant is

$$x^2(x - 2y)^2 - 4x^2y^2 = 0 \text{ or, } x^3(x - 4y) = 0.$$

Hence, $x(x - 4y) = 0$ is the singular solution and $x = 0$ is also tac locus.

8. Reduce the differential equation $(xp - y)(x - py) = 2p$ to Clairaut's form by the substitution $x^2 = u$, $y^2 = v$ and then examine for singular solution.

Solution

From $x^2 = u$, $y^2 = v$, we get $\frac{y dy}{x dx} = \frac{dv}{du}$. Writing $\frac{dv}{du} = P$, we get

$$\left(\frac{x}{y}Px - y\right)\left(x - \frac{x}{y}Py\right) = 2\frac{x}{y}P$$

$$\text{or, } (uP - v)(1 - P) = 2P$$

$$\text{or, } v = uP + \frac{2P}{P-1}$$

which is Clairaut's form and its general solution is

$$v = uc + \frac{2c}{c-1}$$

$$\text{or, } y^2 = x^2c + \frac{2c}{c-1}$$

$$\text{or, } x^2c^2 - c(x^2 + y^2 - 2) + y^2 = 0.$$

The c -discriminant is $(x^2 + y^2 - 2)^2 - 4x^2y^2 = 0$.

Again, the given equation can be expressed as

$$xyp^2 - p(x^2 + y^2 - 2) + xy = 0.$$

The p -discriminant is $(x^2 + y^2 - 2)^2 - 4x^2y^2 = 0$.

Hence the singular solution is $(x^2 + y^2 - 2)^2 - 4x^2y^2 = 0$.

EXERCISE

- Examine for singular solutions (1-18):

1. $4p^2 = 9x$,

2. $x^2 + y = p^2$,

3. $8p^3 = 27y$,

4. $3y = 2px - \frac{2p^2}{x}$,

5. $xp^2 - 2yp + ax = 0$,

6. $4xp^2 = (3x - a)^2$,

7. $y^2(1+p^2) = a^2$, 8. $y + px = x^4 p^2$,
 9. $y = px + e^p$, 10. $y = 2px + p^2$,
 11. $y = px + p^3$, 12. $y = 2px - yp^2$,
 13. $p^2 + xp - y = 0$, 14. $y = px + \frac{a}{p}$,
 15. $y = px - \sqrt{p^2 - 1}$, 16. $y = px + p^3$,
 17. $y = 3px + 6p^2 y^2$, 18. $x^3 p^2 + x^2 yp + 1 = 0$.

• Reduce the following equation to Clairaut's form (19-24) by the substitutions indicated there and solve:

19. $p^4 = 4y(xp - 2y)^2$, put $y = v^2$, $x = u$.
 20. $p^3 - 4xyp + 8y^2 = 0$, put $y = v^2$, $x = u$.
 21. $x^2 p^2 + py(2x + y) + y^2 = 0$, put $y = u$, $xy = v$.
 22. $xp^2 - 2yp + x + 2y$, put $y - x = v$, $x^2 = u$.
 23. $y^2(y - xp) = x^4 p^2$, put $x = \frac{1}{u}$, $y = \frac{1}{v}$.
 24. $(px^2 + y^2)(px + y) = (p + 1)^2$, put $x + y = u$, $xy = v$.
 25. Obtain the differential equation satisfied by the family of curves $c^2 + 2cy - x^2 + 1 = 0$, where c is the parameter. Find also the singular solution.
 26. Obtain the differential equation satisfied by the family of curves $x^2 + y^2 + 2cx + 2c^2 - 1 = 0$, where c is the parameter. Find also the singular solution.

ANSWERS

1. No singular solution, cusp locus $x = 0$.
 2. No singular solution. cusp locus $y + x^2 = 0$.
 3. S.S.: $y = 0$; cusp locus: $y = 0$.
 4. S.S.: $x^3 = 6y$; cusp locus: $x = 0$.
 5. S.S.: $y^2 = ax^2$.
 6. S.S.: $x = 0$, tac locus: $3x - a = 0$, nodal locus: $x - a = 0$.
 7. S.S.: $y = \pm a$; tac locus: $y = 0$.
 8. S.S.: $4x^2 y + 1 = 0$; tac locus: $x = 0$.
 9. S.S.: $y = x(\ln x - 1)$.
 10. No singular solution, cusp locus: $x^2 + y = 0$.
 11. S.S.: $4x^3 + 27y^2 = 0$. 12. S.S.: $y = \pm x$.
 13. S.S.: $x^2 + 4y = 0$. 14. S.S.: $y^2 = 4ax$.

15. S.S.: $y^2 = x^2 - 1$. 16. S.S.: $4x^3 + 27y^2 = 0$.
 17. S.S.: $3x^2 + 8y^3 = 0$. 18. S.S.: $x(xy^2 - 4) = 0$; tac locus: $x = 0$.
 19. G.S.: $y = c^2(x - c)^2$; S.S.: $x^4 = 16y$; tac locus: $y = 0$.
 20. G.S.: $y = c(x - c)^2$; S.S.: $y(27y - 4x^3) = 0$; tac locus: $y = 0$.
 21. G.S.: $xy = cy + c^2$; S.S.: $y(y + 4x) = 0$; tac locus: $y = 0$.
 22. G.S.: $2c^2x^2 - 2c(y - x) + 1 = 0$; S.S.: $x^2 + 2xy - y^2 = 0$.
 23. G.S.: $c^2xy + cy - x = 0$; S.S.: $y(y + 4x^2) = 0$; tac locus: $xy = 0$.
 24. S.S.: $x^2y^2 + 4(x + y) = 0$; tac locus: $y = x$.
 25. $(1 - x^2)p^2 + 2xyp + x^2 = 0$; S.S.: $x^2 + y^2 = 1$; tac locus: $x = 0$.
 26. $2y^2p^2 + 2xyp + x^2 + y^2 - 1 = 0$; S.S.: $x^2 + 2y^2 = 2$; tac locus: $y = 0$.